

Practical Name: DQL & SORTING DATAPractical No. 04

Objective: At the end of the assignments, participants will be able to understand basics of DQL commands, select, conditional retrieval, operators, pattern matching, order by clause.

Theory:

DQL (Data Query Language)

- DQL is used to retrieve data from database tables.
- The SELECT statement is the primary command. It extracts data without modifying the database, making it safe and non-destructive.
- SELECT may be combined with clauses like WHERE, ORDER BY, GROUP BY, etc., to refine output.

Where Clause

- The WHERE clause filters rows in a SELECT statement based on conditions.
- It supports like =, <>, >, <, BETWEEN, IN, LIKE, and IS NULL.
- This enables conditional retrieval and selective extraction of data.

Pattern Matching (LIKE Operator)

- LIKE is used to search patterns in text data.
- Wildcards used:
 - * % : any number of characters
 - * _ : exactly one character
- Example: last name LIKE 'A%' retrieves names starting with A.

ORDER BY Clause

- ORDER BY sorts records in ascending (ASC) or descending (DESC) order.
- It allows sorting based on multiple columns.
- It is always the last clause in a SELECT query.

SUBSTITUTION VARIABLES(&)

- Substitution variables allow runtime input in Oracle SQL using &var and &&var.
- &var prompts every time, whereas &&var prompts only once and reuses the value.

Objective: At the end of this experiment, participants will be able to understand the concept and application of Subqueries and Nested Queries in SQL. We will learn how to retrieve data using single-row and multiple-row subqueries, apply correlated subqueries, and utilise subqueries within SELECT, FROM, and WHERE clauses to efficiently perform complex data retrieval and filtering operations.

Theory:

SUBQUERY

- A subquery is a query inside another SQL query.
- The outer query uses the output of the inner query to perform additional operations.
- Subqueries must be enclosed in parentheses.

TYPES OF SUBQUERIES

(a) Single-Row Subquery:

- Returns exactly one row.
- Uses operators like =, <, >, <=

(b) Multiple-Row Subquery:

- Returns multiple rows.
- Uses IN, ANY, ALL, EXISTS

(c) Correlated Subquery:

- Depends on a value from the outer query to execute & it executes row by row.

Practical Name: SUBQUERY & NESTED QUERY.

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NESTED SUBQUERIES

- A nested subquery is one that contains another subquery inside it.
- It is useful for multi-level filtering.

CORRELATED SUBQUERIES

- These subqueries reference columns from the outer query, making them dependent on the outer result for each evaluation.

SUBQUERY OPERATORS

- * IN: Checks if a value exists in a list returned by the subquery.
- * ANY: Compares with any value from the subquery's result.
- * ALL: Compares with all values from subquery output.
- * EXISTS: True if subquery returns at least one row.